The frozen split of the finite-volume Euler equations on the C-grid

MultiRate notes

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Abstract

For split-explicit (MIS) time integration with a *frozen* right-hand side, $N(q) = F(q, q)$, the first argument y is frozen at the stage values and the second z evolves in the fast loop. This note describes the split of the finite-volume discretization on the Arakawa C-grid (`src/FV/frozen.jl`, flat grids): the potential temperature at the faces and, optionally, the derivative $\mathrm{d}p/\mathrm{d}\Theta$ at the faces are frozen, the momentum advection is the slow part. Two FV schemes are provided, with the pressure gradient ∇p or in product form $(\mathrm{d}p/\mathrm{d}\Theta)\nabla\Theta$; for the product form the analytic $\mathrm{d}p/\mathrm{d}\Theta$ is frozen, the fast part is linear in z and its vertically implicit (HEVI) step is a tridiagonal solve per column. A well-balanced variant without background subtraction, following Artiano et al., is given, and the consequences for the stability of MIS methods are summarized.

1 Equations

Variables ρ, $U = \rho u$, $W = \rho w$, $\Theta = \rho\theta$; the pressure is a function of Θ alone,

$$p = P(\Theta) = p_0\Big(\frac{R\,\Theta}{p_0}\Big)^{\gamma}, \qquad P'(\Theta) = \frac{\gamma p}{\Theta} = \frac{c^2}{\theta}.$$

$$\begin{aligned}
\partial_t\rho + \partial_x U + \partial_z W &= 0,\\
\partial_t\Theta + \partial_x(\theta U) + \partial_z(\theta W) &= 0,\\
\partial_t U + \partial_x(Uu) + \partial_z(Wu) + \partial_x p &= 0,\\
\partial_t W + \partial_x(Uw) + \partial_z(Ww) + \partial_z p + \rho g &= 0,
\end{aligned}$$

plus a Rayleigh sponge on the vertical momentum only, $-\alpha(z)\,W$, near the top (and optionally at the sides). No background state is subtracted (the optional `FV_HYDRO_REFERENCE` subtracts p_{ref}, ρ_{ref} in the momentum equations; it is off by default, and the frozen split rejects such models).

2 Grid and full operator N

Flat C-grid, $n_x \times n_z$ cells of size $\Delta x \times \Delta z$, periodic in x, walls at the bottom and the top. $\rho_{i,k}$ and $\Theta_{i,k}$ at the cell centres, $U_{i-1/2,k}$ at the x-faces, $W_{i,k-1/2}$ at the interior z-faces ($W = 0$ at the walls). Velocities at the faces divide by the arithmetic mean of the two cells, $u_{i-1/2,k} = U_{i-1/2,k}/\{\{\rho\}\}_x$, $w_{i,k-1/2} = W_{i,k-1/2}/\{\{\rho\}\}_z$; $\theta = \Theta/\rho$ at the centres. Differences $\delta_x a_i = (a_{i+1/2} - a_{i-1/2})/\Delta x$ and likewise in z.

Face values. At every face $i + \frac{1}{2}$ a left and a right value are reconstructed from the cell values by the third-order ($\kappa = \frac{1}{3}$) formula,

$$a^L_{i+1/2} = \frac{-a_{i-1} + 5a_i + 2a_{i+1}}{6}, \qquad a^R_{i+1/2} = \frac{2a_i + 5a_{i+1} - a_{i+2}}{6},$$

and the upwind one is used,

$$a_{i+1/2} = \begin{cases} a^L_{i+1/2}, & v \geq 0, \\ a^R_{i+1/2}, & v < 0, \end{cases}$$

v the transporting mass flux at the face (`recon_left`, `recon_right`, `interp3` in `src/FV/advection.jl`). (This is the same value as the third-order upwind-biased flux of WRF, written there as a centred fourth-order mean plus a sign-dependent correction; the reconstruction form makes the left and right states explicit.) In z at the faces next to the walls, where one of the two states lacks its stencil, the centred mean $\frac{1}{2}(a_i + a_{i+1})$ is used.

The operator.

$$\begin{aligned} \partial_t \rho_{i,k} &= -\delta_x U - \delta_z W, \\ \partial_t \Theta_{i,k} &= -\delta_x(\tilde{\theta}_x U) - \delta_z(\tilde{\theta}_z W), \qquad \tilde{\theta}_x,\ \tilde{\theta}_z \text{ the upwind face values of } \theta \text{ for } U,\ W, \\ \partial_t U_{i-1/2,k} &= -A_U - \frac{p_{i,k} - p_{i-1,k}}{\Delta x}, \\ \partial_t W_{i,k-1/2} &= -A_W - \frac{p_{i,k} - p_{i,k-1}}{\Delta z} - g\,\frac{\rho_{i,k-1} + \rho_{i,k}}{2}, \end{aligned}$$

$p_{i,k} = P(\Theta_{i,k})$ (scheme `:gradp`, the default of `FV_PRESSURE_FORM`). The product-form scheme `:product` replaces the pressure differences by

$$p_{i,k} - p_{i-1,k} \;\to\; P'_{i-1/2,k}\,(\Theta_{i,k} - \Theta_{i-1,k}), \qquad P'_{i-1/2,k} = \tfrac{1}{2}\big(P'(\Theta_{i-1,k}) + P'(\Theta_{i,k})\big), \quad P'(\Theta) = \frac{\gamma P(\Theta)}{\Theta},$$

and likewise in z: the derivative is computed analytically in the cells and averaged to the face. It is defined for all neighbouring values (a secant $[\![p]\!]/[\![\Theta]\!]$ is not, for equal values), and its pressure difference differs from $[\![p]\!]$ by $O([\![\Theta]\!]^3)$. Both schemes use the full p and ρ, no background state. The momentum advection A_U, A_W is in flux form with the transporting mass fluxes averaged to the momentum control volumes (e.g. $\bar{U}_{i,k} = \frac{1}{2}(U_{i-1/2,k} + U_{i+1/2,k})$) and the transported velocities as upwind face values (the same reconstruction); A_W uses the centred mean next to the walls.

3 The frozen split

Frozen coefficients $C(y)$, computed once per stage value Y_j (`frozen_data!`):

- $\tilde{\theta}_x(y)$, $\tilde{\theta}_z(y)$: the upwinded face values of θ of the full operator, the upwind direction from U, W of y;
- for the scheme `:product` the face derivatives $P'_f(y)$ (the means of the cell values $\gamma p/\Theta$ of y, as in N).

Fast part (`frozen_fast!`), evolving z:

$$\begin{aligned} \partial_t \rho &= -\delta_x U_z - \delta_z W_z, \\ \partial_t \Theta &= -\delta_x\big(\tilde{\theta}_x(y)\, U_z\big) - \delta_z\big(\tilde{\theta}_z(y)\, W_z\big), \\ \partial_t U &= -\frac{\Pi_{i,k} - \Pi_{i-1,k}}{\Delta x}, \qquad \partial_t W = -\frac{\Pi_{i,k} - \Pi_{i,k-1}}{\Delta z} - g\,\{\{\rho_z\}\}_z, \end{aligned}$$

plus the sponge $-\alpha W$ in the W rows, with the pressure difference across a face

$$\texttt{:gradp}\colon\ \Pi_r - \Pi_l = P(\Theta_{z,r}) - P(\Theta_{z,l}), \qquad \texttt{:product}\colon\ \Pi_r - \Pi_l = P'_f(y)\,\big(\Theta_{z,r} - \Theta_{z,l}\big).$$

Slow part $S(y) = (0, 0, -A_U(y), -A_W(y))$, the momentum advection evaluated at y only.

MIS stage. Stage i integrates

$$\frac{\mathrm{d}Z}{\mathrm{d}\tau} = \sum_j b_{ij}\, F(Y_j, Z) + \sum_j \frac{\gamma_{ij}}{d_i \Delta t}(Y_j - y^n), \qquad b_{ij} = \frac{\beta_{ij}}{d_i}.$$

Since F is affine in the coefficients, $\sum_j b_{ij} F(Y_j, Z) = F_{\bar{C}}(Z) + \sum_j b_{ij} S(Y_j)$ with the weighted coefficients $\bar{C} = \sum_j b_{ij} C(Y_j)$: one fast operator per stage (`frozen_mis_step!`).

4 What is frozen, and what must not be

term	frozen from y	live from z
mass flux	–	U, W
Θ flux	$\tilde{\theta}_x$, $\tilde{\theta}_z$, upwind direction	U, W
pressure, scheme `:gradp`	–	$P(\Theta)$
pressure, scheme `:product`	face means P'_f of $\gamma p/\Theta$	$[\![\Theta]\!]_f$
gravity	–	$\{\{\rho\}\}_z$
momentum advection (slow)	everything	–

Rule of thumb: freeze *coefficients* of the fast operator, never a part of its stiff response. A third option, $p = (p/\Theta)(y)\,\Theta_z$, was tried and removed: since $\partial p/\partial\Theta = \gamma\, p/\Theta$, it puts $(\gamma-1)/\gamma$ of the acoustic stiffness into the frozen argument, and the split is then stable only for $c\,\Delta t/\Delta z \lesssim 1$.

5 Properties

1. $F(q, q) = N(q)$ exactly in both pressure modes (test: relative difference $\lesssim 10^{-10}$).
2. For the scheme `:product` the fast part is linear in z: $F(y, z) = L(y)z + S(y)$, with the coefficients $\tilde{\theta}$, P'_f of y; for `:gradp` it keeps the cellwise $P(\Theta_z)$.
3. Mass and Θ are conserved (flux form), the momentum equations are in flux form as well.
4. The Θ flux of the fast part is the live momentum times the frozen θ, $\tilde{\theta}(y)\,U_z$ and $\tilde{\theta}(y)\,W_z$; it is linear in z, no velocity is formed, and $\tilde{\theta}(y)$ is a coefficient like $P'_f(y)$. The slow part contains no Θ terms. See Section 8.

6 Fast integrators

Explicit. Forward–backward (Verlet) in the thermodynamic rows (ρ, Θ) and the momentum rows, substep limited by the vertical sound speed, $\delta\tau \lesssim \Delta z/c$.

HEVI. Strang splitting: half a horizontal Verlet step, a vertically implicit step, half a horizontal step. The vertical part (vertical fluxes of ρ, Θ, vertical pressure difference, gravity) is advanced by the linearly implicit Crank–Nicolson step $z \leftarrow z + (I - \frac{\delta\tau}{2} J_v)^{-1} \delta\tau\, G_v(z)$. With the pressure difference across the z-face k written as $\alpha_k \Theta_k - \beta_k \Theta_{k-1}$, the elimination of ρ and Θ (as in WRF) gives per column the tridiagonal system for W ($c = \delta\tau/2$, $r = (c/\Delta z)^2$, $\theta_k = \tilde{\theta}_z$ at the face k):

$$W_{k-1}\Big[-r\beta_k\theta_{k-1} + \frac{c^2 g}{2\Delta z}\Big] + W_k\big[1 + r\theta_k(\alpha_k + \beta_k)\big] + W_{k+1}\Big[-r\alpha_k\theta_{k+1} - \frac{c^2 g}{2\Delta z}\Big] = R_k^W - \frac{c}{\Delta z}\big(\alpha_k R_k^\Theta - \beta_k R_{k-1}^\Theta\big) - \frac{c\,g}{2}\big(R_{k-1}^\rho + R_k^\rho\big),$$

then ρ and Θ by back substitution. `:product`: $\alpha_k = \beta_k = P'_{k-1/2}(\bar{y})$, the vertical part is linear and the step is exactly Crank–Nicolson. `:gradp`: $\alpha_k = a_k$, $\beta_k = a_{k-1}$ with $a = \gamma p/\Theta$ at the start of the stage (an approximate Jacobian). The factorization is computed once per stage and step size (`frozen_stage_begin!`, `frozen_vertical_step!`).

7 Well-balanced variant (not implemented)

At a hydrostatic state the W equation leaves the residual $-\big([\![p]\!]_{k-1/2}+g\Delta z\,\{\{\rho\}\}_{k-1/2}\big)/\Delta z$, which vanishes only up to truncation error with the arithmetic mean $\{\{\rho\}\}$. Following Artiano, Knoth, Spichtinger, Ranocha (arXiv:2509.10311), the gravity term is the two-point term $\bar\rho\,[\![\phi]\!]$ ($\phi = gz$, $[\![\phi]\!] = g\Delta z$) with a mean chosen such that $\bar\rho\,[\![\phi]\!] = -[\![p]\!]$ for the hydrostatic state:

$$\text{isothermal: } \bar\rho = \{\{\rho\}\}_{\log} = \frac{[\![\rho]\!]}{[\![\log\rho]\!]}, \qquad \text{constant } \theta: \ \bar\rho = \{\{\rho\}\}_{\gamma} = \frac{\gamma-1}{\gamma}\frac{[\![\rho^\gamma]\!]}{[\![\rho^{\gamma-1}]\!]}.$$

(Isothermal: $[\![\log\rho]\!] = -[\![\phi]\!]/(RT)$, so $\{\{\rho\}\}_{\log}[\![\phi]\!] = -RT[\![\rho]\!] = -[\![p]\!]$; constant θ: $\frac{\gamma}{\gamma-1}K\theta^\gamma\rho^{\gamma-1} + \phi = \text{const}$, so $\{\{\rho\}\}_{\gamma}[\![\phi]\!] = -K\theta^\gamma[\![\rho^\gamma]\!] = -[\![p]\!]$.)

$$\partial_t W_{i,k-1/2} = -A_W - \frac{[\![p]\!]_{k-1/2} + \bar\rho_{k-1/2}\,g\Delta z}{\Delta z}$$

is then exactly balanced for these atmospheres, without any background state. The Θ fluxes vanish at rest ($W = 0$), the horizontal pressure differences vanish for horizontally homogeneous states.

Product scheme. With the product form the hydrostatic balance reads $P'_{k-1/2}[\![\Theta]\!] + \bar\rho\,g\Delta z = 0$. For constant θ ($\Theta = \theta\rho$, $P' = \gamma K\theta^\gamma\rho^{\gamma-1}$, $[\![\rho^{\gamma-1}]\!] = -\frac{\gamma-1}{\gamma K\theta^\gamma}\,g\Delta z$) the balancing mean is

$$\bar\rho = (\gamma-1)\,\frac{\{\{\rho^{\gamma-1}\}\}\,[\![\rho]\!]}{[\![\rho^{\gamma-1}]\!]} \quad (\to \rho \text{ for equal values}).$$

Frozen form. Freeze, from y, the face derivative $P'_{k-1/2}$ and the ratio $R_{k-1/2} = \bar\rho_{k-1/2}/\{\{\rho\}\}_{k-1/2}$, and take $[\![\Theta_z]\!]$ and $\{\{\rho_z\}\}$ live:

$$\partial_t W = -\frac{P'_{k-1/2}(y)\,[\![\Theta_z]\!]_{k-1/2} + R_{k-1/2}(y)\,\{\{\rho_z\}\}_{k-1/2}\,g\Delta z}{\Delta z}.$$

The fast part stays linear, $F(q,q) = N(q)$, and the column system above only changes its gravity coefficients ($g \to R_{k-1/2}\,g$ in the two $c^2g/(2\Delta z)$ entries and in the R^ρ terms). For a general stratification (constant Brunt–Väisälä frequency in the gravity-wave tests) neither mean is exact; $\{\{\rho\}\}_\gamma$ is the natural default for the troposphere.

8 Stability of MIS methods for this split

For a linear problem the frozen split with exact fast stages is MIS with $L = \partial F/\partial z$ and $N - L = \partial F/\partial y$. Linearized about a hydrostatic state with uniform wind $\bar u$, both operators are invariant under shifts of the columns, and one horizontal Fourier mode gives a block over the whole column (`MethodDesign/app_stability.jl`). On the gravity-wave grid ($\Delta x = 2.5\,\text{km}$, $n_z = 16$) the methods that are stable on the one-dimensional acoustic-advection test of Knoth and Wensch grow by 5–25 % per step already from $C_{\text{adv}} = \bar u\Delta t/\Delta x \approx 0.4$. In the frozen split the advection of θ enters only through the frozen coefficient $\tilde\theta(y)$ of the Θ flux (Section 5, item 4); this structure is not present in that test. Methods designed on the column blocks (`MethodDesign/search_app.jl`):

method	frozen order	$\max(\rho-1)$, $C_{\text{adv}} \le 1.6$, $\bar u = 10/20/40$ m/s
MIS54 (Bauer & Knoth)	2	0.16 / 0.18 / 0.10
MISF53 (MIS54 projected onto frozen order 3)	3	0.14 / 0.26 / 0.26
MIS_FV42 (designed, from MIS4)	2	$7\cdot10^{-4}$ / $7\cdot10^{-3}$ / $7\cdot10^{-4}$
MIS_FV53 (designed, from MISF53)	3	$\le 6\cdot10^{-4}$

Nonlinear gravity wave (300 km channel, $U = 20\,\mathrm{m/s}$, `:hevi`): at $T = 3000\,\mathrm{s}$ all are accurate (MIS_FV53: wave error $1.2 \cdot 10^{-4}\,\mathrm{K}$ at $\Delta t = 130\,\mathrm{s}$, $1.8 \cdot 10^{-4}\,\mathrm{K}$ at $200\,\mathrm{s}$; MIS54: $1.6 \cdot 10^{-4}$, $1.6 \cdot 10^{-4}$); in a long run (36000 s at $\Delta t = 200\,\mathrm{s}$, $C_{\mathrm{adv}} = 1.6$) only MIS_FV53 survives. The frozen order (the bicoloured B-series conditions, `MethodDesign/BiColourMIS.jl`) is 2 for MIS54 and 3 for MISF53 and MIS_FV53.

9 3D and cut cells

The module `MultiRate.FV3D` (`src/FV3D`) extends the split to three dimensions: ρ, Θ at the centres, U, V, W on the faces, periodic in x and y. One grid type serves flat grids and Cartesian cut cells: V_f the fluid volume fraction of a cell, A_x, A_y, A_z the open fractions of its faces (from a terrain $z_s(x, y)$ by quadrature, or exactly for a box obstacle); $1/V_{\mathrm{eff}} = 1/\max(V_f, \alpha_{\min})$ in the divisions. The fast part per unit fluid volume:

$$\begin{aligned}
\partial_t \rho &= -\tfrac{1}{V_{\mathrm{eff}}}\big[\delta_x(A_x U) + \delta_y(A_y V) + \delta_z(A_z W)\big],\\
\partial_t \Theta &= -\tfrac{1}{V_{\mathrm{eff}}}\big[\delta_x(A_x \tilde\theta_x(y) U) + \delta_y(A_y \tilde\theta_y(y) V) + \delta_z(A_z \tilde\theta_z(y) W)\big],\\
\partial_t U &= -\delta_x p, \quad \partial_t V = -\delta_y p, \quad \partial_t W = -\delta_z p - g\{\{\rho\}\}_z \qquad \text{(open faces)},
\end{aligned}$$

with the same pressure forms as in 2D, and the column system of Section 6 generalizes with $s_k = c/(\Delta z\, V_{\mathrm{eff},k})$ and the apertures A_k:

$$\begin{aligned}
&W_{k-1}\big[-r\beta_k s_{k-1} A_{k-1}\theta_{k-1} + \tfrac{cg}{2} s_{k-1} A_{k-1}\big] + W_k\big[1 + r A_k \theta_k(\alpha_k s_k + \beta_k s_{k-1}) + \tfrac{cg}{2} A_k (s_k - s_{k-1})\big]\\
&\quad + W_{k+1}\big[-r\alpha_k s_k A_{k+1}\theta_{k+1} - \tfrac{cg}{2} s_k A_{k+1}\big] = R^W_k - r(\alpha_k R^\Theta_k - \beta_k R^\Theta_{k-1}) - \tfrac{cg}{2}(R^\rho_{k-1} + R^\rho_k).
\end{aligned}$$

The operator is written as building blocks (thermodynamic and momentum part, horizontal and vertical, forcing, sponge); the parts of the fast loops are compositions of them, and $N(q) = F(q, q)$ is evaluated with the same blocks. For $n_y = 1$ the flat 3D operator is the 2D one to rounding. No background state is subtracted; a discretely balanced initial column (`hydrostatic_column`) is exactly at rest, also with obstacles, and gives second-order convergence of the frozen MIS step (from the analytic profile the order drops to ≈ 0.8). Flow over a hill and over a cube aligned with the grid or shifted into cut cells: stable, conservative to rounding, the same wave field for both cubes (FINDINGS §13).

10 Open points

- The split is implemented on flat grids only; terrain (Gal-Chen metric terms in the pressure gradient) and cut cells need the frozen coefficients of their metric and aperture terms.
- The well-balanced gravity mean of Section 7 is not implemented.
- Consistent advection: freezing the transported θ splits the Θ advection between the fast and the slow part; a split that keeps the advection of ρ and Θ together (e.g. also freezing the advective part of the mass flux) would remove the coupling that limits MIS stability.
- The DG counterpart: `docs/frozen_dg/frozen_dg.tex`.